

Similarity Detection

Parfois

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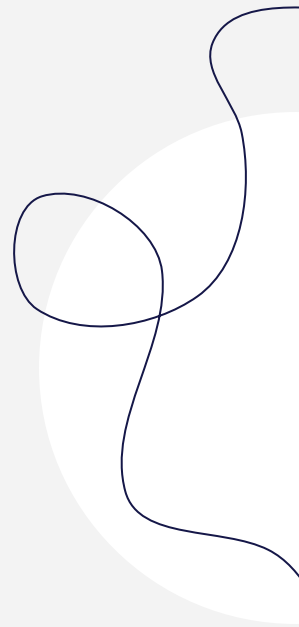
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01

Exploratory Data Analysis





1. Exploratory Data Analysis

Remove duplicate values

- There's only one image per product and color.
- We removed the several sizes for products of one specific color (by keeping the first record in `df_product`).

Missing products in `df_sales`

- `df_product` has 386 more instances than the `df_sales`.

Handling Missing Values

- We dropped features with more than 75% of missing values, as they have too little information to be considered.

Columns with only one value

- We removed columns that only presented one unique value.

Keep only one ID column

- Only keep the feature that identifies the pictures.

Remove similar columns

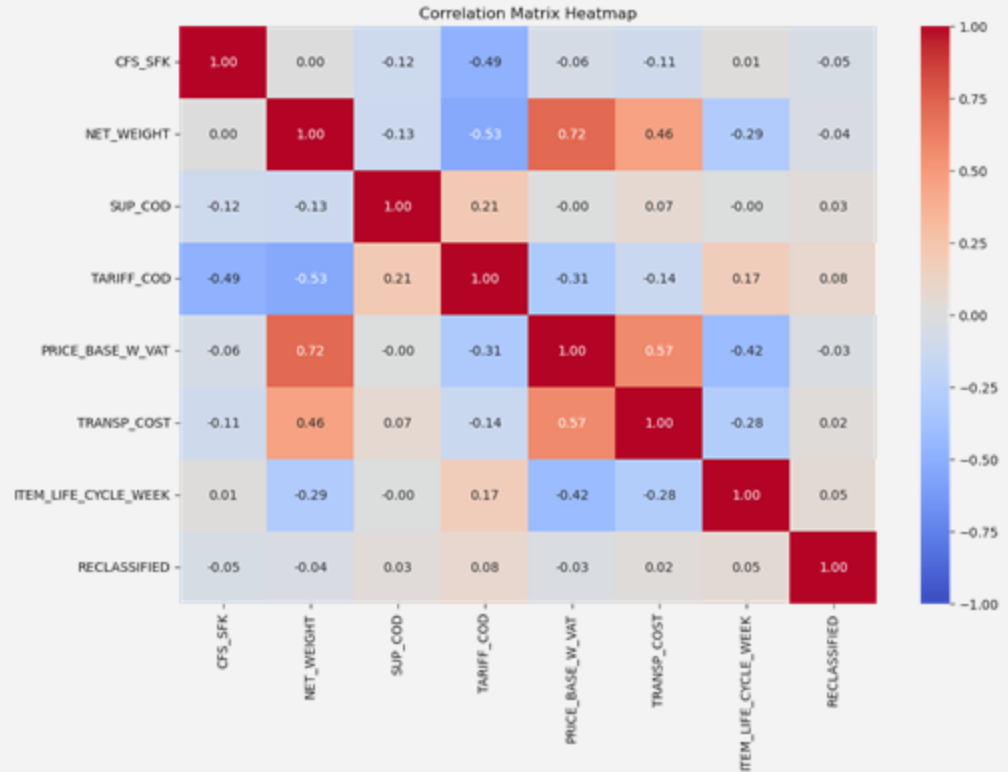
- Find pairs of similar columns using Jaccard Similarity (with a threshold = 0.9), and keep only one of them.
- Identify features that have the same prefix/suffix, and compare their information.



1. Exploratory Data Analysis

Correlation between numerical variables

- No correlation above 0.75 (so no variables that “contain the same information”).
- No numerical columns removed.



02

Preprocessing



2. Preprocessing

Image Preprocessing

- Most of the images have a white background.
- However, some of the images, have noise in the background:
 - Models showcasing the products;
 - Shadows, text and other noise.

Since noise dominates the images, it significantly affects the process of identifying similar products.



Image preprocessing: Segmentation



2. Preprocessing

Image Preprocessing

Image semi-automated segmentation model in three steps:

1. Identification of images with noise;
2. Selection of region to segment;
3. Image segmentation using Segment Anything Model (SAM)



03

Modeling

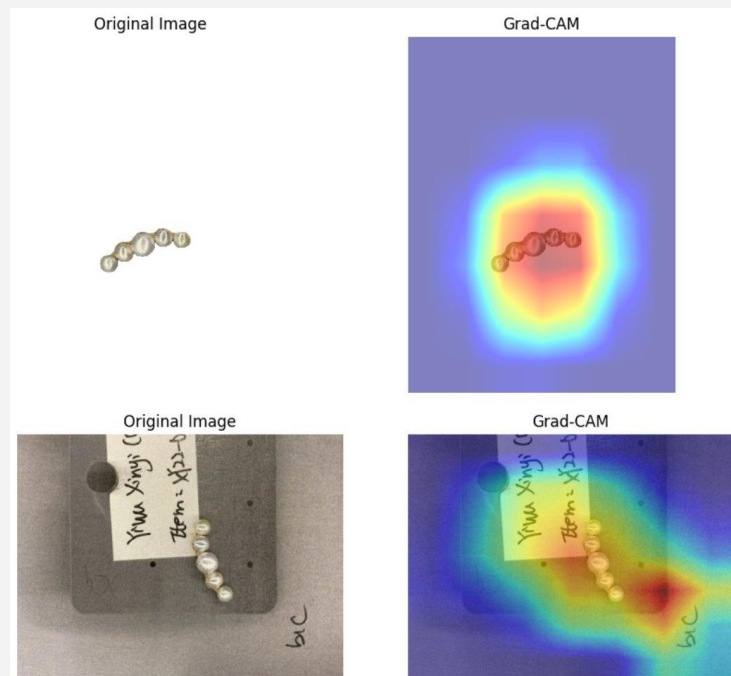
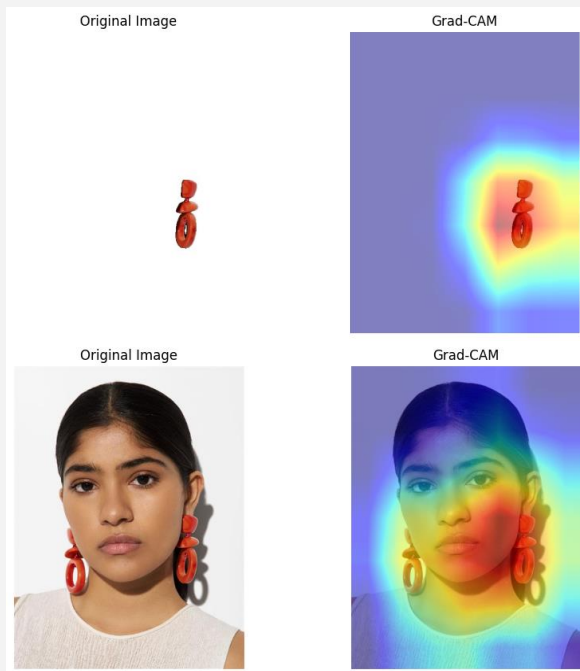




ResNet for Image Segmentation

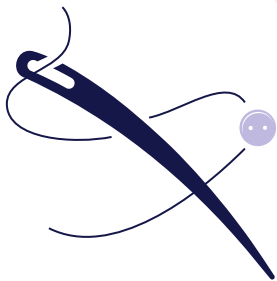
- Pre-trained ResNet50 model to extract features
 - Removal of fully connected layers
 - Use **penultimate layer** as general representations of images
- Grad-CAM to interpret embeddings
 - To check the regions ResNet focused on

Grad-CAM



XGBoost for Products Data Segmentation

- Self-supervised XGBoost feature extractor model:
 - Input features are used as both the input and the target
 - Model captures relationships and patterns within the dataset



Feature	Weight	Gain
CATEGORY_MATRIX	11.34%	0.41%
COMPOSITION	10.40%	89.38%
MAIN_THEME	9.67%	8.77%
CLR_DES	9.29%	0.21%
PRICE_BASE_W_VAT	8.33%	0.07%
GFS_DES_EN	7.49%	0.30%
DISPLAY_DES	5.10%	0.03%
GFA_DES_EN	5.10%	0.15%
STORE_MONTH_FINAL_PC	3.88%	0.01%
FASHIONTYPE	3.14%	0.01%
PRICE_OUTLET_W_VAT	2.89%	0.08%
FINISHING	2.84%	0.06%
COLLECTION_TYPE	2.65%	0.01%
PROD_SEG	2.31%	0.00%
PRINT_TYPE	2.02%	0.00%
L4_DES	1.99%	0.28%
MATERIAL	1.89%	0.04%
NUMBER_OF_UNITS	1.82%	0.05%
L3_DES	1.73%	0.03%
DIMENSION	1.40%	0.01%
DISTRIBUTION_BLOCK	1.40%	0.01%
FSALDOS	1.27%	0.01%
CATEGORY_MATRIX_BUYER	0.79%	0.03%
CAT_DES_EN	0.58%	0.00%
COLLECTION_TYPE_GROUP	0.39%	0.01%
L2_DES	0.15%	0.02%
L1_DES	0.15%	0.02%

04

Recommendations



Explainability of Results

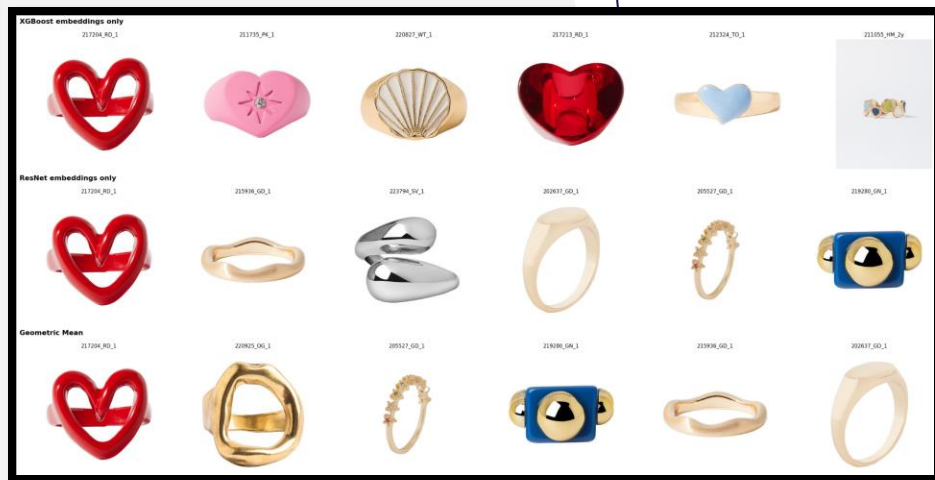
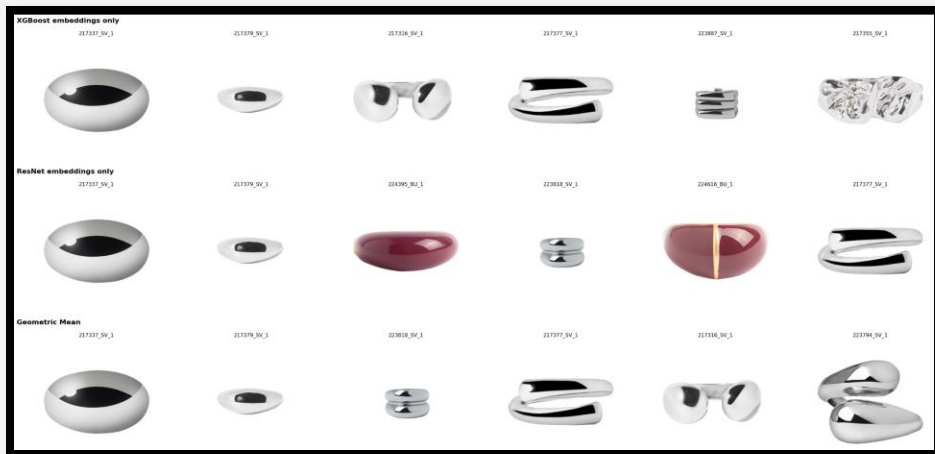
- filter by GFA_DES_EN category was explicitly added to the function of finding similar products



PROG_IMAGE	Similarity_Score	CATEGORY_MATRIX	COMPOSITION	MAIN_THEME	CLR_DES	PRICE_BASE_W_VAT	GFS_DES_EN	DISPLAY_DES	GFA_DES_EN	STORE_MONTH_FINAL_PC	FASHIONTYPE	PRICE_OUTLET_W_VAT	FINISHING
217337_SV_1	1.000000	52 - Rings	Zinc	SHADES OF COOL	Silver	5.99	Band Ring	Panel	Rings	January	Basic	3.99	Rodhium
217379_SV_1	0.999629	52 - Rings	Zinc	SHADES OF COOL	Silver	4.99	Band Ring	Panel	Rings	January	Basic	2.99	Rodhium
217316_SV_1	0.999629	52 - Rings	Zinc	SHADES OF COOL	Silver	4.99	Band Ring	Panel	Rings	January	Basic	2.99	Rodhium
217377_SV_1	0.998984	52 - Rings	Zinc	SHADES OF COOL	Silver	5.99	Cocktail	Panel	Rings	January	Basic	3.99	Rodhium
223887_SV_1	0.998238	52 - Rings	Zinc	SILVER BASICS	Silver	6.99	Band Ring	Panel	Rings	July	Basic	3.99	Rodhium
217355_SV_1	0.997861	52 - Rings	Zinc	SHADES OF COOL	Silver	5.99	Fantasy	Panel	Rings	January	Basic	3.99	Rodhium

Combinations of Embeddings

- Calculate cosine similarity matrices for embeddings
- Normalize each matrix using global min and max
- Calculate geometric mean of 2 matrices



Final Results

Geometric Mean

216081_HM_1



217561_GN_1



207912_BG_1



217616_BM_1



216928_HM_1



218918_PK_1



207912_DM_1



215097_DM_1



216832_HM_1



212122_SK_1



211212_BG_1



217246_BM_1



206255_BM_1



217357_GN_1



215613_BG_1



219781_GY_1



212496_HM_1



211832_SK_1



207912_NUJ_1



214097_BM_1



218921_HM_1



214682_BU_1



222304_BG_1



215097_BM_1



05

Sales Forecasting





Sales Forecast

- To forecast the sales of the selected product, we calculated a **weighted mean** of the sales of the most similar products. The **weights** are the similarity scores extracted from the model (normalized so that their sum equals 1).

Product						
Real Sales (Qty)	2270.0	1091.0	3636.0	4335.0	1866.0	1430.0
Similarity Score (normalized)	–	0.2004	0.2004	0.1999	0.1997	0.1996
Predicted Sales	2471.795					



06

Conclusion + Q&A





Conclusion

Image Quality

The presence of visually consistent, plain background images was critical for the performance of the model. Noisy images were tackled using Segmentation.

Different approaches attempted

We found that a combination of XGBoost and Resnet embeddings, using geometric mean, provided the best results.

Model Evaluation

A labeled test set would allow a more accurate measure of the model's performance.



Thanks!

Do you have any questions?

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